# MVP Simulator: MDP Formulation and Initial Results

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## 1 MDP Formulation

The MVP simulator implements a finite-horizon MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$:

### 1.1 State Space

$$\mathcal{S} = \{\texttt{sedentary},\ \texttt{active}\}$$

Two activity states with no additional time-of-day dimension. Time-dependent reward scaling is handled via `reward_multiplier_by_step` (see §1.4 below).

### 1.2 Action Space

$$\mathcal{A} = \{\texttt{nudge},\ \texttt{idle}\}$$

A binary intervention: send a motivational message (`nudge`) or do not (`idle`).

### 1.3 Transition Function

$P(s' \mid s, a)$ is a configurable $2 \times 2 \times 2$ tensor. Rows index current state $s \in \{\text{sedentary}, \text{active}\}$ and columns index next state $s'$ in the same order:

$$P_{\text{nudge}} = \begin{bmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{bmatrix} \qquad P_{\text{idle}} = \begin{bmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{bmatrix}$$

Sending a reminder increases $P(\text{active} \mid \text{sedentary})$ from 0.1 to 0.3 but decreases $P(\text{active} \mid \text{active})$ from 0.6 to 0.5, consistent with Reactance theory (reminders can backfire for already-active users).

Optional `reward_multiplier_by_step`: a per-step scaling factor that can zero out rewards at certain daily steps (soft mask). When absent, all multipliers default to 1.0. The per-step reward is computed as base reward $\times$ multiplier$[t]$ where $t$ is the step-of-day index.

### 1.4 Reward

The base reward is a vector indexed by next state:

$$r = [0.0,\ 1.0]^\top \quad (\text{sedentary},\ \text{active})$$

The per-step reward at time $t$ scales by the multiplier:

$$r(s, a, t) = m[t \bmod S] \cdot r(s')$$

where $S =$ `steps_per_day`, $m \in \mathbb{R}^S$ is `reward_multiplier_by_step` (default $m_i = 1.0$), and $s'$ is the next state after taking action $a$ in state $s$. Currently $m_i = 1.0$ for all $i$, so $r(s, a, t) = r(s')$.

Single-timescale placeholder. Multi-timescale reward (immediate + delayed body measure) is deferred to Phase 2.

### 1.5 Episode Structure

$T =$ `episode_days` $\times$ `steps_per_day` timesteps. Default: $T = 90$ days $\times$ 5 steps/day $= 450$.

## 2 Agents

All agents are bandits: they do not condition on state. From the transition matrix, $E[r \mid \text{nudge}] \approx 0.375$, $E[r \mid \text{idle}] = 0.20$, and a state-aware policy that nudges only when sedentary would achieve $\approx 0.429$. This gap motivates the state-aware agents planned for Phase 2.

### 2.1 Thompson Sampling

Beta-Bernoulli TS with prior Beta$(1, 1)$ per action. Posterior update:

$$\alpha_a \leftarrow \alpha_a + \mathbb{1}[s'.\texttt{activity} = \texttt{active}], \quad \beta_a \leftarrow \beta_a + \mathbb{1}[s'.\texttt{activity} = \texttt{sedentary}]$$

Action selected by sampling $\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$ and choosing $a^* = \arg\max_a \theta_a$.

### 2.2 Epsilon-Greedy

$\epsilon = 0.1$. With probability $\epsilon$, select uniformly at random; otherwise select $\arg\max_a Q(a)$. Q-values updated via incremental average:

$$Q(a) \leftarrow Q(a) + \frac{r_t - Q(a)}{n_a}$$

### 2.3 UCB1

Upper Confidence Bound with exploration parameter $c = 2.0$. Selects:

$$a^* = \arg\max_a \left[ Q(a) + c\sqrt{\frac{\ln N}{N_a}} \right]$$

where $N$ is total steps and $N_a$ is the number of times action $a$ was chosen. The confidence bonus shrinks as $N_a$ grows, naturally shifting from exploration to exploitation. Each action is pulled once before UCB is applied.

### 2.4 Random Baseline

Uniform random action selection. No learning. Serves as lower bound. The mixture chain $P_{\text{random}} = \frac{1}{2}(P_{\text{nudge}} + P_{\text{idle}})$ gives $E[r] \approx 0.308$ per step.

## 3 Initial Results

All agents run over 10 random seeds (1–10), 450 timesteps each. Results are reported as mean $\pm$ standard deviation across seeds.

### 3.1 Standard Config (`config/rule_based.yaml`)

No `reward_multiplier_by_step` – uniform 1.0 multiplier on all steps.

| Agent | Total Reward | Mean Reward/Step |
|---|---|---|
| Thompson Sampling ($\alpha_0 = \beta_0 = 1$) | 160.3 ± 10.5 | 0.356 ± 0.023 |
| Epsilon-Greedy ($\epsilon = 0.1$) | 160.8 ± 15.3 | 0.357 ± 0.034 |
| UCB ($c = 2.0$) | 148.2 ± 15.0 | 0.329 ± 0.033 |
| Random | 135.6 ± 10.7 | 0.301 ± 0.024 |

Table 1: Results on `config/rule_based.yaml` (10 seeds, 450 timesteps). TS and $\epsilon$-greedy perform similarly; TS has lower variance. UCB outperforms random but shows higher variance.

### 3.2 Masked Config (`config/rule_based_with_mask.yaml`)

`reward_multiplier_by_step: [1, 1, 1, 1, 0]` – step 4 (end of day) yields zero reward regardless of outcome. All agents drop proportionally to the masked fraction (1/5 of steps zeroed), confirming the multiplicative reward model.

| Agent | Total Reward | Mean Reward/Step |
|---|---|---|
| Thompson Sampling ($\alpha_0 = \beta_0 = 1$) | 126.6 ± 6.8 | 0.281 ± 0.015 |
| Epsilon-Greedy ($\epsilon = 0.1$) | 127.3 ± 10.2 | 0.283 ± 0.023 |
| UCB ($c = 2.0$) | 116.9 ± 8.6 | 0.260 ± 0.019 |
| Random | 107.8 ± 6.7 | 0.240 ± 0.015 |

Table 2: Results on `config/rule_based_with_mask.yaml` (10 seeds, 450 timesteps). Reward drops $\approx 20\%$ across all agents, matching the 1/5 masked fraction. Relative ordering is preserved.

## 4 Known Limitations

- Agents are state-blind bandits, not state-aware RL. State-aware agents are Phase 2.
- Single-timescale reward only. Multi-timescale (immediate + 3-week delayed) is Phase 2.
- No user response model. Burden accumulation and 4 archetypes are Phase 2.
- Transition matrix is fixed and known. Learned transitions from real data are Phase 2.
- Transition probabilities are placeholder values informed by Reactance theory. Calibration against real data is Phase 2.